Bank Marketing Campaign Analysis Report

Term Deposit Subscription Analysis

Introduction

Banks frequently run direct marketing campaigns to promote financial products such as term deposits, most commonly through outbound telephone calls to existing and prospective customers. While telemarketing remains an effective way to reach customers, it can be costly and time-consuming when a large share of contacts are unlikely to subscribe. This report analyzes a bank marketing campaign dataset of 11,162 client contacts, comparing customers who subscribed to a term deposit against those who declined, in order to identify the customer characteristics and campaign factors most closely associated with successful subscriptions.

The report is organized into the following sections: business problem statement, dataset description, data analysis process, key findings, business insights, recommendations, and conclusion.

Business Problem Statement

The bank commits considerable financial and human resources to direct marketing campaigns, yet more than half of contacted customers (52.62%) decline the term deposit offer. This results in low conversion rates and unnecessary operating costs, particularly where customers are contacted repeatedly with little chance of success. The central business challenge addressed in this report is to identify which customer characteristics and campaign factors most strongly influence whether a customer subscribes to a term deposit. Understanding these factors allows the bank to target customers more effectively, reduce wasted contact effort, and improve overall campaign profitability.

Dataset Description

The dataset contains customer demographic information, financial attributes, and marketing campaign details. Variables include age, job, marital status, education, credit default status, account balance, housing loan, personal loan, contact type, contact day and month, call duration, number of campaign contacts, days since previous contact, previous contacts, previous campaign outcome, and the target variable 'deposit', indicating whether the customer subscribed to a term deposit. Both numerical and categorical variables make the dataset suitable for descriptive analysis, visualization, and predictive modeling.

The dataset includes the following variables:

- **age:** Customer's age.
- **job:** Occupation of the customer.
- **marital:** Marital status.
- **education:** Highest education level.
- **default:** Credit default status.
- **balance:** Average yearly account balance.
- **housing:** Whether the customer has a housing loan.
- **loan:** Whether the customer has a personal loan.
- **contact:** Contact communication type.
- **day:** Day of the month the customer was contacted.
- **month:** Month of contact.
- **duration:** Duration of the last phone call (seconds).
- **campaign:** Number of contacts made during the current campaign.
- **pdays:** Days since the customer was last contacted.
- **previous:** Number of contacts before the current campaign.
- **poutcome:** Outcome of the previous marketing campaign.
- **deposit:** Target variable indicating subscription to a term deposit.

Data Analysis Process

The analysis began with the bank marketing dataset, which was cleaned and organized into two comparative dashboards, one for subscribers and one for non-subscribers, built in Power BI. Data quality checks were performed to identify inconsistencies and missing values in categorical fields such as job, education, and previous campaign outcome. Descriptive statistics were calculated for key numerical variables, including call duration, account balance, and age, and compared across both customer groups.

Categorical variables such as marital status, education level, occupation, housing loan, personal loan, contact method, and previous campaign outcome were visualized using bar charts, donut charts, and pie charts to identify patterns in subscription behavior. The resulting dashboards and summary tables formed the basis for the key findings, business insights, and recommendations presented in this report.

Key Findings

Customers who subscribed to a term deposit spent considerably more time on the phone with the bank, averaging 537 seconds per call compared to 223 seconds for customers who declined, a difference of more than double. This suggests that meaningful, extended conversations play an important role in successful conversions.

Financial profile also distinguished the two groups. Subscribers held higher average account balances ($1,804 versus $1,280) and carried less existing debt, with 63% holding no housing loan compared to 43% of non-subscribers, and 91% holding no personal loan compared to 83% of non-subscribers.

Previous campaign outcome was informative but incomplete, as “unknown” was the largest category for both groups, likely a mix of genuinely new customers and records that were never properly logged. Among customers with a known outcome, subscribers were far more likely to have a previous successful outcome, while non-subscribers were dominated by previous failures.

Occupation also shaped subscription behavior. Management professionals formed the largest subscriber group, while blue-collar workers made up 21.05% of non-subscribers compared to only 13.39% of subscribers, the largest gap of any occupation category.

Married customers and those with secondary education represented the largest subscriber groups, and cellular contact was the most effective communication channel, reaching 82.6% of subscribers compared to 62.5% of non-subscribers. Finally, contact volume peaked in May for both groups, though non-subscriber contacts were more concentrated within a small number of months.

Business Insights

These findings indicate that targeted marketing outperforms contacting customers indiscriminately. Customers with stronger financial profiles, lower existing debt, and a history of responding positively to previous campaigns represent the bank's most promising prospects.

The quality and length of customer interactions appear to matter more than the number of contact attempts, suggesting that call center performance should be measured on engagement rather than volume alone. Incomplete previous-outcome records also point to a data quality gap that, if addressed, would allow the bank to distinguish new customers from customers whose campaign history was simply not captured, improving the accuracy of future targeting efforts.

Recommendations

Based on these insights, the following actions are recommended:

- Segment customers before launching campaigns, rather than applying a single strategy to the entire customer base.
- Develop tailored messaging for blue-collar customers, who are currently under-represented among subscribers relative to their share of total contacts.

- Prioritize call quality over call volume by training marketing representatives to sustain longer, more engaged conversations.
- Re-target customers with a previous successful campaign outcome, as this is the strongest repeat-conversion signal identified in the data.
- Improve data capture procedures so that "unknown" previous-outcome records are minimized, distinguishing genuinely new customers from incomplete records.
- Align staffing and campaign scheduling with seasonal contact patterns, particularly the peak observed in May.

Conclusion

This analysis of the bank marketing campaign dataset demonstrates that subscription to a term deposit is influenced by a combination of financial profile, existing debt, occupation, demographics, communication channel, and the quality of engagement during the call itself. Applying these insights allows the bank to move from a broad, one-size-fits-all marketing approach toward a targeted, evidence-based strategy that improves conversion rates while reducing wasted contact effort. Continued investment in data quality, customer segmentation, and performance monitoring will support sustained improvements in marketing efficiency and profitability.

Appendix: Detailed Dashboard Analysis

This appendix presents the full breakdown of the subscriber and non-subscriber dashboards underlying the findings above.

A.1 Headline Metrics

Metric	Subscribers	Non-Subscribers
Total Contacts	5,289	5,873
Conversion / Decline Rate	47.38%	52.62%
Avg. Call Duration	537.18 sec	223.13 sec
Avg. Balance	$1,804	$1,280
Avg. Age	42	41

Total dataset: 11,162 client contacts.

A.2 Previous Outcome

Both groups are dominated by "Unknown" (no prior campaign record), but the shape differs. Subscribers: Unknown (largest), then Success, then Failure, then Other — Success is a meaningful second-place category. Non-Subscribers: Unknown (largest), then Failure, then

Other, then Success (smallest) — Success is barely present. A documented past success is strongly tied to subscribing; a documented past failure is strongly tied to declining.

A.3 Marital Status

Both groups follow the same order, Married > Single > Divorced, but married customers make up a larger share of the subscriber base, and non-subscribers have a noticeably larger single population relative to married.

A.4 Contact Method

Channel	Subscribers	Non-Subscribers
Cellular	4.37K (82.6%)	3.67K (62.54%)
Unknown	0.53K (10.0%)	~1.82K (~30.9%)
Telephone	~0.39K (~7.4%)	0.38K (6.54%)

Non-subscribers have roughly three times more "unknown" contact records than subscribers, a reachability and data-quality gap rather than simply a channel-preference difference.

A.5 Loan Status

Personal Loan:

	Subscribers	Non-Subscribers
No	4.81K (90.85%)	4.9K (83.38%)
Yes	0.48K (9.15%)	0.98K (16.62%)

Housing Loan:

	Subscribers	Non-Subscribers
No	3.35K (63.41%)	2.53K (43.03%)
Yes	1.94K (36.59%)	3.35K (56.97%)

Both loan types show the same direction: subscribers carry meaningfully less debt, especially housing loans (a 20-point gap versus an 8-point gap for personal loans).

A.6 Level of Education

Both groups peak at Secondary, followed by Tertiary, then Primary, with Unknown smallest. The distribution shape is similar across both groups; education alone does not strongly separate subscribers from non-subscribers, though secondary-educated customers are the largest subscriber segment in absolute terms.

A.7 Job / Occupation

Job	Subscribers %	Non-Subscribers %
Management	24.60%	21.54%
Technician	15.88%	16.74%
Blue-Collar	13.39%	21.05%
Admin.	11.93%	11.97%
Retired	9.76%	4.46%
Services	6.98%	9.43%
Student	5.09%	1.55%
Unemployed	3.82%	2.64%
Self-Employed	3.54%	3.71%
Entrepreneur	2.33%	3.49%
Housemaid	2.06%	2.81%
Unknown	0.64%	0.61%

The biggest gaps are Blue-Collar (over-represented among non-subscribers by roughly 7.7 points) and Retired (over-represented among subscribers by roughly 5.3 points).

A.8 Monthly Contact Volume

Month	Subscribers %	Non-Subscribers %
May	17.49%	32.33%
Aug	13.01%	14.15%
Jul	11.85%	15.10%
Apr	10.91%	5.89%
Jun	10.32%	11.51%
Feb	8.34%	5.70%
Nov	7.62%	9.19%
Oct	6.11%	1.17%
Sep	5.09%	0.85%
Mar	4.69%	0.48%
Jan	2.68%	3.44%
Dec	1.89%	0.17%

May dominates both groups, but far more heavily for non-subscribers, who received 32% of all their contacts in that one month. Subscriber volume is spread more evenly across the year, especially in September and October, where non-subscriber volume nearly disappears.

A.9 Synthesis

The dashboard tells a consistent story: subscribers are reached more reliably (cellular contact, fewer unknown records), engage longer on the call, carry less debt, skew toward management and retired occupations, and often have a documented past success. Non-subscribers cluster heavily into a single high-volume month (May), suggesting a broad, less targeted pattern of contact rather than a focused one.